Dilemma Dynamics

Cooperation, deliberation and repeated-game incentives

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Abstract

Cooperation can reward patience, but it can also be a costly mistake. Does an AI that drafts before acting better recognise which situation it faces? Dilemma Dynamics studies this question in repeated Prisoner's Dilemma games, where the immediate gain from defection competes with the future value of reciprocity. We compare immediate action with a bounded private draft in NVIDIA's Nemotron Nano and Super at two sampling temperatures. Against disclosed opponent policies, exact dynamic programming measures the payoff given up at each choice. Autonomous pairings reveal how cooperation begins, breaks down and returns. Matched game schedules, interleaved temperatures and durable event logs connect aggregate comparisons to individual decisions. The final evaluation is underway; these notes develop its theoretical motivation, experimental design and reproducible analysis.

1 The question and the intervention

The experiment asks whether a short deliberation procedure helps a model respond to an opponent's incentives. Its two hosted models are Nemotron Nano and Super [6, 7]. The exact provider IDs, under the `nvidia/` namespace, are `NVIDIA-Nemotron-3-Nano-30B-A3B` and `nemotron-3-super-120b-a12b`.

Direct makes one thinking-off action request, capped at 128 completion tokens. *Draft* makes a thinking-on request capped at 1,024 tokens, then a separate thinking-off action request capped at 128 tokens [8]. Both receive completed game history. Only the current decision's draft is added to its action request. Each episode is assigned a sampling temperature $\tau \in \{0, 1\}$; all its request phases use that value. The nucleus-sampling setting is fixed at top-$p = 1$. Each model–procedure combination is an experimental *arm*.

Players maximise expected own cumulative points. Choices remain sealed until joint reveal. Neutral X/Y labels use both mappings and display seats. In canonical action labels, C means cooperate and D means defect:

Own action	Other cooperates	Other defects
Cooperate	$(3, 3)$	$(0, 5)$
Defect	$(5, 0)$	$(1, 1)$

Each pair lists (own points, other points). Write $R = 3$ for the mutual-cooperation reward, $S = 0$ for cooperating against defection, $T = 5$ for defecting against cooperation and $P = 1$ for mutual defection. These satisfy $T > R > P > S$.

2 Related literature and theoretical foundations

Rasmusen's *Games and Information* [1] supplies the organising ideas: strategies specify responses to histories, best responses maximise payoff against a given opponent, and equilibrium requires compatible best responses. Its treatment of finite and infinite repetition explains how the prospect of another encounter changes present incentives. Its discussions of incomplete information and the Axelrod tournament also motivate studying simple commitments alongside adaptive players. Here those ideas lead to two complementary tracks: exact payoff comparisons against disclosed policies, and observed interaction between models whose policies emerge during play.

Akata et al. [2] study finitely repeated games with language models, programmed strategies and human players. Their social chain-of-thought procedure encourages a model to anticipate the other player's behaviour and improves coordination in their experiments. This provides a close precedent for comparing decision procedures. Dilemma Dynamics uses a bounded draft with a separate final action request, making the procedure's token limits and request sequence explicit.

Fontana, Pierri and Aiello [3] examine Llama2, Llama3 and GPT3.5 in 100-round Prisoner's Dilemma interactions against random opponents with different cooperation probabilities. They check understanding of the rules and history and find substantial variation in responses to opponent hostility. Their work motivates our engineering checks and opponent-specific reporting. The present benchmark adds an exact own-payoff reference for each disclosed commitment, allowing cooperation and decision quality to be examined separately.

Pal et al. [4] infer strategies from decisions across possible recent histories, vary continuation probabilities, payoffs, horizons and framing, and assess equilibrium properties. Their approach emphasises the value of observing responses to histories that ordinary play may rarely visit. Our autonomous track records realised trajectories; its cooperation, breakdown and recovery statistics describe the interaction paths actually sampled.

Taylor and Bergen [5] study action–message incongruence in games with a communication phase. They vary the incentives and timing of messages and report associations between reasoning performance and misrepresentation. This places the present intervention in a wider behavioural context: payoff competence, cooperation and truthful communication are distinct properties of an agent. Our sealed-action design concentrates on the first two.

Together, these studies motivate a question about *when* deliberation helps: a change in average score may combine better exploitation of an unresponsive opponent, better preservation of reciprocity and different behaviour under uncertainty. The disclosed-policy track exposes those components, while matched temperature comparisons test their sensitivity to the sampling setting.

3 Repetition and the value of a future encounter

An *episode* is one complete interaction between two players. Round number is $t = 1, 2, \ldots$, and N denotes the total number of rounds in that episode. In a fixed-horizon episode, $N = H$, where the positive integer H is announced before play. In a geometric episode, continuation probability δ satisfies $0 \leq \delta < 1$. After each completed round, the game continues with probability δ and ends with probability $1 - \delta$, independently of the actions.

One equivalent construction draws an independent uniform variable $U_t \sim \text{Uniform}(0, 1)$ after round t, continuing precisely when

$$U_t < \delta.$$

The realised length then has distribution and mean

$$\mathbb{P}(N = n) = (1 - \delta)\delta^{n-1}, \qquad \mathbb{P}(N \geq t) = \delta^{t-1}, \qquad \mathbb{E}[N] = \frac{1}{1 - \delta}, \quad n, t \geq 1. \tag{1}$$

For $\delta = 0$, the game lasts one round. The first survival probability is one. Thus $\delta = .9$ produces ten expected rounds, with a hidden last round; an announced ten-round horizon gives players different information. The executable scheduler uses seeded, exact rational draws for the specified probabilities. Matched treatments reuse the same environment draws.

3.1 Committed responses

The disclosed opponents are AllC (always C), AllD (always D), Tit-for-Tat (C first, then the subject's preceding action) and Grim (C until a subject defection, then D forever). This makes the consequences of a current action computable. Rasmusen's distinction between a commitment and strategic equilibrium is useful here [1, Sections 4.1 and 5.1–5.2].

Against untriggered Grim, cooperation forever has value $R/(1 - \delta)$. Defecting now, followed by the best response to permanent punishment, has value $T + \delta P/(1 - \delta)$. Cooperation is weakly preferable exactly when

$$\frac{R}{1 - \delta} \geq T + \frac{\delta P}{1 - \delta} \quad \Longleftrightarrow \quad \delta \geq \frac{T - R}{T - P} = \frac{1}{2}. \tag{2}$$

All four payoff symbols were defined with the matrix; δ is the chance of another round. Equality allows both initial choices to be optimal.

Against initially cooperating Tit-for-Tat, three continuation patterns determine the optimal initial value: cooperate forever, defect forever, or alternate D,C. Their respective values are

$$\frac{3}{1 - \delta}, \qquad 5 + \frac{\delta}{1 - \delta}, \qquad \frac{5}{1 - \delta^2}.$$

The alternating pattern earns 5,0 repeatedly. Comparing all three shows that initial cooperation is optimal for $\delta \geq 2/3$. The tested probabilities .4, .6 and .9 therefore span different incentive regimes.

4 Exact payoff references and decision loss

Track A tests each disclosed policy at all three continuation probabilities. One-round AllC/AllD and announced ten-round Tit-for-Tat/Grim complete 16 conditions. For any condition, let s be the opponent's relevant memory state, $a \in \{\mathrm{C}, \mathrm{D}\}$ the subject's action, $r(s,a)$ the resulting own payoff and $f(s,a)$ the next opponent state. Let $V^*(s)$ be the maximum achievable expected total own payoff from state s; $Q^*(s,a)$ fixes the current action and allows optimal choices thereafter. In a geometric episode,

$$Q^*(s,a) = r(s,a) + \delta V^*(f(s,a)), \qquad V^*(s) = \max_{a \in \{\mathrm{C},\mathrm{D}\}} Q^*(s,a). \tag{3}$$

The implementation uses rational arithmetic. It certifies a candidate value function by checking $V^*(s) \geq r(s,a) + \delta V^*(f(s,a))$ for every state and action, with equality for at least one action in each state.

For a finite game, let h be rounds remaining, including the current one. Backward induction starts at $V_0^*(s) = 0$ and applies

$$Q_h^*(s,a) = r(s,a) + V_{h-1}^*(f(s,a)), \qquad V_h^*(s) = \max_a Q_h^*(s,a).$$

Against committed ten-round Tit-for-Tat or Grim, the optimal initial path is nine C moves followed by D, earning $9R+T = 32$ points. The standard finite-game backward-induction argument in which both players remain free to optimise uses a different opponent model [1, Section 5.1].

At a revealed decision, s_t is the pre-action opponent state and a_t the chosen action. Its nonnegative payoff gap is

$$g_t = V^*(s_t) - Q^*(s_t, a_t),$$

with the corresponding remaining-horizon values in a finite episode. Episode losses are

$$L_{\text{geometric}} = (1-\delta)\sum_{t=1}^{N} g_t, \qquad L_{\text{fixed}} = \frac{1}{H}\sum_{t=1}^{H} g_t. \tag{4}$$

Zero gap means that the current action preserves optimal achievable payoff. For example, C against AllC in a one-round game gives up $T - R = 2$ points; D has gap zero. Loss is expressed in normalised own-payoff points.

Equation (1) explains the geometric weighting: survival to round t already occurs with probability δ^{t-1}. The expectation of the realised sum in (4) therefore includes that survival factor. Applying it again to each recorded gap would count stopping twice.

Track B tests all ten unordered pairings of the four arms, including self-play, at $\delta = .9$. Here outcomes of interest are cooperation and joint return. For player $i \in \{1,2\}$, write $r_{i,t}$ for points in round t. A dyad's normalised joint return is $(1-\delta)\sum_{t=1}^{N}(r_{1,t} + r_{2,t})$. We report first-round mutual cooperation, cooperation per observed round and per episode, and breakdown/recovery counts with opportunity denominators. Long episodes receive more weight in a pooled round rate; rounds in one dyad share a history and are statistically dependent.

5 Replication and statistical comparisons

A block groups the related experimental conditions and arms. Each block has 64 Track A episodes (16 conditions for each of four arms) and ten Track B dyads per temperature. The frozen resource-only selection chose 48 blocks at each temperature: 6,144 benchmark episodes and 960 autonomous dyads, or 7,104 episodes in total. Four strata, defined by action-label mapping and display seat, each contain 12 blocks per temperature.

A fresh master seed fixes game schedules and order. Arms within a block share environment draws, and corresponding temperature episodes are adjacent in a frozen randomised order. This pairing controls stopping and presentation variation while spreading both temperatures through the same service window. Nested 24/32/48-block candidates were frozen before engineering; measured token use and latency selected the largest fitting candidate before scientific inference. Selection uses resource telemetry alone.

5.1 Primary drafting effects

Write $L_{m,k,\tau,b,c}$ for Track A loss in model $m \in \{\text{Nano}, \text{Super}\}$, procedure $k \in \{\text{Direct}, \text{Draft}\}$, temperature $\tau \in \{0, 1\}$, block b and condition $c \in \{1, \ldots, 16\}$. The block's condition mean is

$$\bar{L}_{m,k,\tau,b} = \frac{1}{16} \sum_{c=1}^{16} L_{m,k,\tau,b,c}.$$

For the contrast under consideration, let $\mathcal{B}_j$ be the set of blocks in stratum $j \in \{1, 2, 3, 4\}$ with complete observations for both procedures; $|\mathcal{B}_j|$ denotes its size. The drafting effect is

$$\Delta_{m,\tau} = \frac{1}{4} \sum_{j=1}^{4} \frac{1}{|\mathcal{B}_j|} \sum_{b \in \mathcal{B}_j} \left(\bar{L}_{m,\text{Draft},\tau,b} - \bar{L}_{m,\text{Direct},\tau,b} \right). \tag{5}$$

Negative values favour Draft. All conditions and strata have equal weight. The completed design has $|\mathcal{B}_j| = 12$ in every stratum. The point estimate requires at least one complete paired block in every stratum.

There are four primary contrasts, one for each model and temperature. Use 50,000 deterministic PCG64 bootstrap replicates, resampling whole blocks within strata. For family error level $\alpha = .05$ and $K = 4$ contrasts, each marginal interval has nominal coverage $1 - \alpha/K = .9875$. Its bootstrap quantiles are $\alpha/(2K) = .00625$ and $1 - \alpha/(2K) = .99375$. This Bonferroni allocation uses approximate bootstrap coverage. At least five complete paired blocks per stratum are required to construct an interval.

5.2 Temperature comparisons and missingness

Six secondary contrasts compare temperature 0 minus 1 for each of the four arms, and $\Delta_{m,0} - \Delta_{m,1}$ for each model. A positive value of the latter means a smaller drafting advantage at zero. Resampling uses the same matched block indices jointly across temperatures; secondary 95% intervals are nominal.

Every scheduled episode retains a status. Estimates from complete blocks are conditional on their completion. Dynamic programming bounds feasible loss over the remaining scheduled rounds, given any committed prefix. These bounds describe the finite schedule's missingness exposure. The release gate requires completion of both temperatures before the public report and app change over.

6 Evidence, execution and interpretation

The local CPU coordinates remote Nebius Token Factory inference. Inspect AI provides native samples, model events and custom scored outcomes [9]. The audited gateway owns HTTP transmission, reservations and recovery; native joint-reveal events carry hashes linking them to the canonical event chain. MLflow Skinny records run identities, parameters, metrics and artifacts in local SQLite [10]. NumPy supplies the bootstrap, PyArrow numeric Parquet exports and Matplotlib the primary comparison figure.

The evidence packet retains append-only requests and revealed rounds, native Inspect logs, accounting, statuses and exports. Offline reconstruction produces episode histories, decision/request tables, condition means, contrasts and checksums. The dependency lock and runtime manifest record the installed state; source archives preserve the executed implementation. Private responses and drafts remain local.

6.1 Bounded recovery

The additional campaign ceiling is $40: $2 engineering, $34 final inference, $2 other-charge reserve and $2 contingency. Retry exposure is included in stage allocations. Reservations precede transmission, and ambiguous requests retain their exposure. Durable valid responses are replayed. A successful but invalid response stays in the record and ends its admitted scientific observation.

A provider-overload interruption led to an explicitly approved transport amendment: six total attempts per logical request, one durable 30-minute recovery window per decision, a shared five-minute overload cooldown, and two sample workers sharing four request slots. The original campaign deadline remains 36 hours, with its watchdog and shutdown margin. The original source and ledger prefix remain intact; a separate frozen receipt records the amendment.

6.2 Reading the eventual findings

The primary effects compare complete bounded procedures: Draft adds computation, a request, thinking controls and conditioning. Opponent breakdowns explain which incentive regimes contribute to an average effect. Autonomous pairings add a separate view of the cooperation paths those procedures generate.

Interpretation remains specific to the tested models, prompts, budgets and hosted service window. Backend weights and sampling may change even when the requested model name stays fixed, including at temperature zero. Bootstrap intervals have approximate coverage; secondary and exploratory comparisons need corresponding caution. Draft-content review supports qualitative questions about the generated text, while behavioural conclusions use revealed actions and exact scores. Broader claims about model rankings, latent beliefs or learning outcomes would require additional experiments.

The public report, replay and downloads will use the completed verified final campaign together. Publication checks require native sample coverage, resolved final reservations, MLflow export and reproducible derived artifacts. `FINAL_EVALUATION.md` gives the operational commands and source identities; `AGENTS.md` records the execution safeguards.

A Probabilities and the app's forecast inequality

The optional app protocol asks for an opponent-action forecast and two conditional future-payoff forecasts. These are elicited separately from the final evaluation's action-only scientific requests. Their algebra helps explain the difference between immediate temptation and future incentives.

Let $\mathcal{H}_t$ be the completed history available before round t, and let $p_t \in [0,1]$ be the reported probability that the opponent cooperates in that round. Thus p_t concerns the opponent's *current action*; δ concerns whether a *later round exists*. The simultaneous protocol keeps the opponent's current choice sealed while the subject chooses.

Let $F_{a,t}$ be the reported expected own total from the next round onward, conditional on continuation and on choosing $a \in \{\mathrm{C}, \mathrm{D}\}$ now. It includes future opponent responses and subsequent stopping. Whenever no further round is possible, define $F_{a,t} = 0$. The probability of reaching the next round is

$$\rho_t = \begin{cases} \delta, & \text{geometric game,} \\ 1, & \text{fixed game with } t < H, \\ 0, & \text{fixed game with } t = H. \end{cases}$$

The conditional future totals have bounds

$$0 \le F_{a,t} \le \begin{cases} 5/(1-\delta), & \text{geometric game,} \\ 5(H-t), & \text{fixed game,} \end{cases}$$

because five is the maximum own payoff per round. In the geometric case, conditioning on the next round means the remaining expected length is again $1/(1-\delta)$.

Write $\widehat{Q}_t(a)$ for expected own payoff according to these submitted forecasts. With the payoff matrix in Section 1,

$$\begin{aligned} \widehat{Q}_t(\mathrm{C}) &= p_t R + (1-p_t)S + \rho_t F_{\mathrm{C},t} = 3p_t + \rho_t F_{\mathrm{C},t}, \\ \widehat{Q}_t(\mathrm{D}) &= p_t T + (1-p_t)P + \rho_t F_{\mathrm{D},t} = 1 + 4p_t + \rho_t F_{\mathrm{D},t}. \end{aligned}$$

Cooperation is weakly preferred according to the forecasts precisely when

$$\rho_t (F_{\mathrm{C},t} - F_{\mathrm{D},t}) \ge 1 + p_t. \tag{6}$$

The right side is the immediate gain from defection, between one and two points; the left side is cooperation's forecast future advantage, weighted by the chance of reaching that future. Equality means a tie. At an announced final round, $\rho_t = 0$, so the immediate incentive favours D for every $p_t \in [0,1]$.

This check measures consistency between an action and submitted forecasts. Forecast accuracy can then be studied against outcomes that were actually observed. A one-step counterfactual holds the other player's sealed current action fixed; a longer alternative history requires specifying how that opponent would respond.

B Competitive board-game calculations

The app's visible sequential board games provide additional examples of tactics and exact search. They use separate rules and analyses from the cooperation study.

B.1 Tic-tac-toe

Let x denote a board state including the player to move, $A(x)$ its legal moves and $f(x, a)$ the state after move a. Let $G(x)$ be optimal game value from the current player's perspective. Terminal values are win $+1$, draw 0 and loss -1, according to that same perspective. Because each move passes the turn, a nonterminal state satisfies

$$G(x) = \max_{a \in A(x)} [-G(f(x, a))].$$

The game is small enough for exhaustive minimax. Legality, search value and the player's explanation can therefore be checked separately.

B.2 Dots & Boxes

Here x includes drawn edges, box ownership and the player to move. Write $g(x, a)$ for the number of boxes captured by drawing edge a, and $W(x)$ for optimal remaining-box difference from the current player's perspective. An edge that captures a box grants another turn. Thus

$$W(x) = \max_{a \in A(x)} \left[g(x, a) + \sigma(x, a) W(f(x, a)) \right], \qquad \sigma(x, a) = \begin{cases} +1, & g(x, a) > 0, \\ -1, & g(x, a) = 0. \end{cases}$$

At a terminal state, $W(x) = 0$, since no boxes remain. The difference in boxes already owned is added separately to obtain the final score difference. The positive sign preserves perspective after an extra turn; the negative sign switches it when the turn passes.

Exact edge search is used within explicit state and computation limits, with incomplete searches labelled unknown. Restricted long-chain/loop endgames also admit specialised references. A box sacrifice can preserve control and increase the eventual score; its value depends on the remaining position and the validity of that reference's assumptions.

Both board games have competitive outcomes: a gain in the score difference is a loss for the other player. The repeated dilemma adds the possibility that future mutual cooperation improves both players' returns. This makes it useful to present tactical foresight and cooperation incentives as separate examples.

References

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